# Performance Trade-offs in Static and Dynamic Thread Pool Architectures

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## 1 Introduction & Motivation

The advancements in multicore architecture and multiprocessor computers are increasing and ongoing [1]. Modern concurrent systems have to choose between static thread pools (pre-created threads) and dynamic thread pools (threads created on demand), but they lack the empirical data to make informed decisions. Static thread pools waste memory during low load and idle periods which results in a constant overhead regardless of the workload. Dynamic thread pools avoid this waste by creating threads only when needed, but they incur overhead for thread creation and destruction, which can lead to performance degradation under high load. In this project, we want to evaluate the performance trade-offs between static and dynamic thread pools under varying workloads and system configurations and answer the question: At what point does the overhead of dynamic thread creation outweigh its memory savings compared to static thread pools?

This tradeoff can directly impact the performance and costs of cloud-based systems [3]. A web service handling variable traffic has to decide between a static pool with constant resource usage or a dynamic pool that adapts to demand but may suffer from latency spikes during high traffic. As cloud providers charge based on resource usage, understanding this tradeoff can lead to significant cost savings while maintaining performance. Apart from costs, this question also affects other aspects of system design such as database connection pools, task scheduling in distributed systems, etc which all use thread pools to manage concurrency. Yet, there is a lack of comprehensive studies that quantify the performance implications of static vs dynamic thread pools across different workloads and system configurations. By systematically evaluating these trade-offs, we can provide practical guidelines for system architects to choose the right thread pool strategy based on their specific workload and performance requirements.

## 2 Background & Related Work

Existing research mainly focuses on optimizing thread pool sizes rather than comparing different thread pool architectures. Current heuristics that are used to determine the number of threads are flawed [7]. In our preliminary research, we find that there have been few researches on dynamic model for thread size. One of the first such model was the Watermark model which adjusts the number of threads according to the current number of incoming requests [5]. The issue with this model is that while it solves the problem of "efficient usage of resources", it doesn't efficiently obtain the required number of threads in advance [4]. Furthermore, there are also prediction based models to determine thread pool size, but in such models, only factors such as the request rate or number of worker threads is considered in the management [6]. One of the research that has proposed dynamic thread pool model(prediction based), themselves point out that the model lacks testing for a longer period of time and on general server environment which provides various services simultaneously [7]. Lee et al. [6] adds to the prediction model by proposing what they call the "Trendy exponential moving average model" and again the issue with this model seems to be that it doesn't have the desired accuracy in its prediction.

A comprehensive study comparing the tradeoffs between static and dynamic thread pools under varying workloads and system configurations is missing. This tradeoff can directly impact the performance and costs of cloud-based systems [3]. Freire et al. [3] compare global thread pools (single pool and queue for all tasks) against local thread pools (each task has its own dedicated pool and queue). Their study simulates these configurations under high workloads to demonstrate that optimized local pools result in a lower average makespan when compared to global pools. While Freire et al. [3] provide insights into thread pool configurations, they focus on optimal distribution of a fixed number of threads, while our work aims to compare static vs dynamic thread pool architectures across varying workloads and system settings. By systematically evaluating these trade-offs, we can provide practical guidelines for system architects to choose the right thread pool strategy based on their specific workload and performance requirements.

## 3 Challenges

When implementing two different thread pools to compare their performance, one challenge is to make sure that the comparison is fair. We need to make sure that both implementations are the same in terms of how they execute instructions and the only difference is how they manage the threads. As shown by Desrochers [2], that the throughput gap between a lock-free queue and lock_based queues can range from 14x to 200x depending on the platform. We need to make sure that both implementations use the same data structures such as queues, locks, etc and the same algorithms to manage to schedule the tasks and threads.

# 4 Proposed Solution

We address the shortcomings mentioned in 2 by implementing minimal static and dynamic thread pool architectures in C++ that differ only in their thread lifecycle management policies, while sharing the same task scheduling and execution logic. The static thread pool will pre-create a fixed number of threads at initialization, which will remain active throughout the lifetime of the pool. The dynamic thread pool will start with zero threads and will create new threads on-demand when tasks arrive. Idle threads will be terminated after a configurable timeout period to free up resources. Importantly, both implementations will use the same task queue, task scheduling algorithm (e.g., FIFO), and other shared components to ensure a fair comparison. This approach will allow us to isolate architectural differences from quality unlike comparing off the shelf implementations like Intel TBB or Boost, which may have optimizations that effect performance independently of thread pool policy.

## 4.1 Evaluation

To evaluate our thread pool implementations, we benchmarked them under three traffic patterns, each defined by a task arrival process that reflects different traffic scenarios:

- Steady load: a constant arrival rate of 1000 tasks/second for the full duration of the run.
- Burst Load: a burst of 2000 tasks/second followed by a 2-second idle period, repeated in cycles.
- Ramping Load: alternating phases on 1500 tasks/second and 500 tasks/second in repeated cycles.

To isolate how task characteristics interact with thread-pool policy, we will apply two controlled micro-benchmarks:

- Memory-based tasks: $N \times N$ matrix multiplications, where $N$ is tunable to scale memory pressure.
- Latency-based tasks: a synthetic task that blocks for a configurable sleep duration $T$ to emulate downstream I/O or service latency.

For each task type, we evaluate three benchmark intensities: Light, Medium, and Heavy. For latency-based tasks, each request performs a blocking sleep with $T \in \{100\,\mu s, 1\,ms, 10\,ms\}$ to represent Light, Medium, and Heavy service times, respectively. For memory-based tasks, we use $N \times N$ matrix multiplication with $N \in \{128, 256, 512\}$ to represent Light, Medium, and Heavy working-set sizes, respectively. For fairness, we fixed the CPU resources available to the benchmark by restricting it to exactly 10 cores using Linux CPU affinity (e.g., tasket).

We evaluated the following experiment matrix (Table 1):

| Dimension | Values |
|---|---|
| Traffic Patterns | Steady load, Burst Load, Ramping Load |
| Task Types | Memory-based, Latency-based |
| Task Intensities | Light, Medium, Heavy |
| Pool Types | Static, Dynamic |
| Core Counts | 10 cores |

**Table 1: Experiment configuration matrix.**

For each task arrival pattern, we measured throughput (tasks completed per second), latency (p50, p95, p99 time from task submission to completion), PSS Memory (Proportional Set Size attributable RAM used by the process), and thread count (active threads over time). We fixd the CPU resources available to the benchmark by restricting 10 cores. For each configuration, we run the benchmark for 120 seconds and repeat the process three times. we will then aggregate the metrics across the three trials to account for system noise and variability.

# 5 Results

As referenced in section 4.1, we ran a total of 36 configurations (3 traffic patterns × 2 task types × 3 task intensities × 2 pool types) with three trials each, resulting in 108 benchmark runs. For these experiments, we set the static thread pool to have a fixed size of 10 threads, matching the number of CPU cores that were pinned. For the dynamic thread pool, we set a minimum thread count of 0 and a rate of spawning a new thread for every 10 tasks in the queue, with the maximum set to OS limits. We did not set a max thread count for the dynamic thread pool as at a cap the dynamic pool would behave like the static pool and we wanted to observe the effects of over-scaling.

## 5.1 Latency-Based Tasks

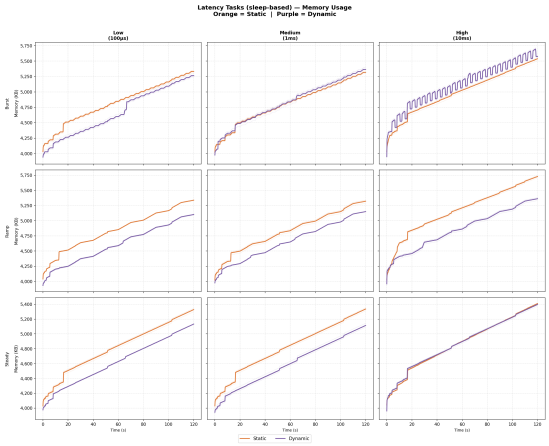
Table 2 reports mean throughput, latency percentiles, memory, and thread count for all latency-based configurations averaged over three trials. Figure 1 shows the corresponding time-series traces.

| Traffic | Task | Pool | Tput (t/s) | p99 (ms) | Mem (MB) | Threads |
|---|---|---|---|---|---|---|
| Burst | Light | Static | 998.9 | 0.20 | 4.7 | 12.0 |
| | | Dynamic | 998.9 | 0.20 | 4.6 | 4.0 |
| | Medium | Static | 998.9 | 1.10 | 4.7 | 12.0 |
| | | Dynamic | 998.9 | 1.10 | 4.7 | 4.7 |
| | Heavy | Static | 992.3 | 2591.75 | 4.9 | 12.0 |
| | | Dynamic | 998.9 | 10.10 | 5.0 | 15.5 |
| Ramp | Light | Static | 998.9 | 0.20 | 4.7 | 12.0 |
| | | Dynamic | 998.9 | 0.20 | 4.5 | 4.9 |
| | Medium | Static | 998.9 | 1.11 | 4.7 | 12.0 |
| | | Dynamic | 998.9 | 1.10 | 4.6 | 5.2 |
| | Heavy | Static | 992.3 | 5617.00 | 5.0 | 12.0 |
| | | Dynamic | 998.9 | 10.11 | 4.7 | 19.2 |
| Steady | Light | Static | 998.9 | 0.19 | 4.7 | 12.0 |
| | | Dynamic | 998.9 | 0.19 | 4.5 | 4.5 |
| | Medium | Static | 998.9 | 1.09 | 4.7 | 12.0 |
| | | Dynamic | 998.9 | 1.10 | 4.5 | 4.9 |
| | Heavy | Static | 992.3 | 865.42 | 4.8 | 12.0 |
| | | Dynamic | 998.9 | 10.10 | 4.8 | 13.4 |

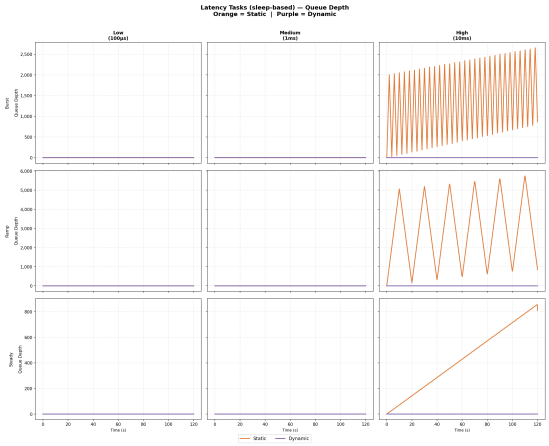
**Table 2: Mean results across three trials for latency-based tasks. Mem is mean PSS in MB; Threads is mean active thread count.**

## 5.2 Matrix Multiplication Tasks

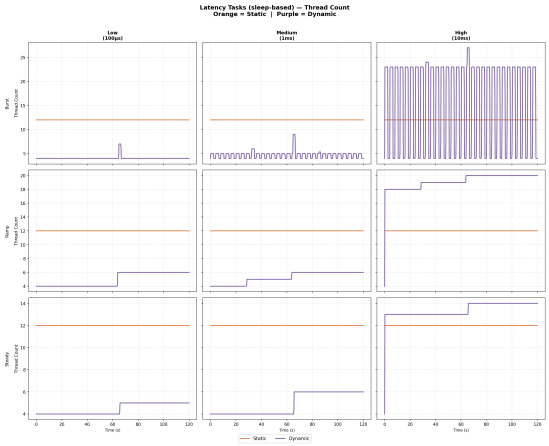
Table 3 reports mean throughput, latency percentiles, memory, and thread count for all matrix multiplication configurations averaged




**(a) Memory usage (PSS)**




**(b) Queue depth over time**




**(c) Active thread count over time**

**Figure 1: System metrics for latency-based tasks across all traffic patterns and intensities.**

over three trials. Figure 2 shows the corresponding time-series traces.

| Traffic | $N$ | Pool | Tput (t/s) | p99 (ms) | Mem (MB) | Threads |
|---|---|---|---|---|---|---|
| Burst | 128 | Static | 999.0 | 5.74 | 7.8 | 12.0 |
| | | Dynamic | 998.9 | 6.62 | 16.5 | 17.9 |
| | 256 | Static | 628.5 | 71463.92 | 21.1 | 12.0 |
| | | Dynamic | 252.0 | 7267.56 | 546.7 | 23420.4 |
| | 512 | Static | 65.8 | 1687830.86 | 70.4 | 12.0 |
| | | Dynamic | 62.2 | 48941.92 | 4707.7 | 7125.4 |
| Ramp | 128 | Static | 998.9 | 6.56 | 8.0 | 12.0 |
| | | Dynamic | 998.9 | 6.59 | 8.6 | 14.0 |
| | 256 | Static | 628.5 | 72535.67 | 21.1 | 12.0 |
| | | Dynamic | 253.3 | 44085.01 | 2042.2 | 137086.4 |
| | 512 | Static | 65.9 | 1685673.80 | 70.4 | 12.0 |
| | | Dynamic | 60.4 | 151306.84 | 7050.5 | 43557.9 |
| Steady | 128 | Static | 998.9 | 6.60 | 7.9 | 12.0 |
| | | Dynamic | 998.8 | 6.55 | 7.3 | 10.0 |
| | 256 | Static | 628.0 | 70277.60 | 21.0 | 12.0 |
| | | Dynamic | 185.1 | 38719.17 | 2105.8 | 150466.3 |
| | 512 | Static | 65.8 | 1685253.26 | 70.4 | 12.0 |
| | | Dynamic | 55.1 | 841820.59 | 7221.8 | 133053.4 |

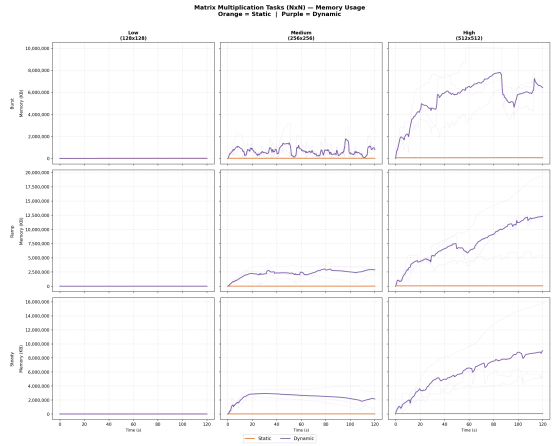
**Table 3: Mean results across three trials for matrix multiplication tasks ($N \times N$ matrices). Mem is mean PSS in MB; Threads is mean active thread count.**

## 6 Conclusion

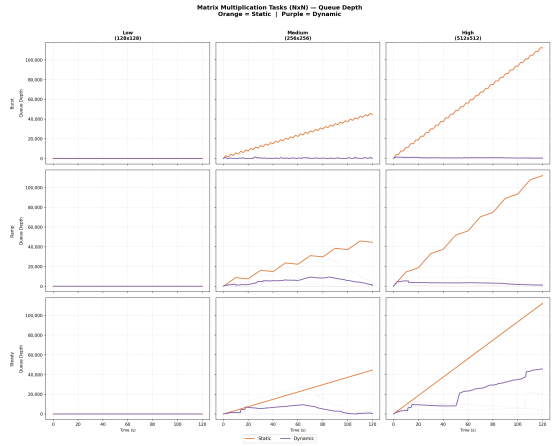
In this project, we aim to systematically evaluate the performance trade-offs between static and dynamic thread pool architectures under varying workloads and traffic patterns. We can summarize our findings as follows:

- Dynamic pool wins for I/O-bound / latency-sensitive workloads : thread scaling prevents queue buildup when tasks can block; the heavier the task and the tighter the arrival rate, the larger the advantage
- Static pool wins for CPU-bound workloads: a fixed thread count matching available cores avoids context-switching overhead, cache thrashing, and memory bloat from excess threads.
- Thread over-scaling is harmful under compute pressure: spawning threads beyond core count produces diminishing returns then actively hurts throughput; memory from concurrent large allocations can exceed system RAM and trigger swapping
- Traffic pattern matters only when the pool is near saturation : Burst and Ramp expose queuing instability in the static pool under heavy latency tasks, but have negligible effect when throughput is the bottleneck (e.g. large matrix multiplication)

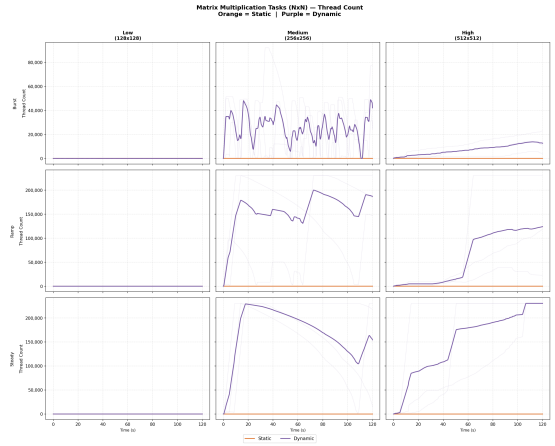
Overall, our results suggest that the choice between static and dynamic thread pools should be primarily guided by the workload characteristics and how much active work the threads are expected to do.




**(a) Memory usage (PSS)**




**(b) Queue depth over time**




**(c) Active thread count over time**

**Figure 2: System metrics for matrix multiplication tasks across all traffic patterns and intensities.**

# References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